

# Context-Aware Vulnerability Prioritization for Government Endpoint Fleets: Integrating Exploit Intelligence, Asset Criticality, and Endpoint Telemetry

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**Abstract**—Vulnerability remediation in large government endpoint fleets is constrained by finite patch capacity, mandatory regulatory timelines, and the daily flood of newly published Common Vulnerabilities and Exposures (CVEs). Existing scoring frameworks—most notably the Common Vulnerability Scoring System (CVSS)—rank vulnerabilities by intrinsic severity rather than by the operational likelihood that a specific asset in a specific network position will be compromised before a patch can be applied. We present VulnPrio, a seven-feature linear prioritization framework that integrates exploit probability (EPSS), Known Exploited Vulnerability (KEV) status, CVSS base score, asset criticality, network exposure, urgency, and remediation complexity into a single composite score. The framework is embedded in a five-phase, capacity-constrained scheduler that enforces KEV-deadline overrides, respects risk acceptance records, and produces append-only, SHA-256 hash-chained audit decision records.

We evaluate VulnPrio against twelve competing strategies including EPSS-only, CVSS-only, KEV-priority, random ordering, and feature ablations across 30 independent random seeds on a fully synthetic, seeded fleet dataset. The primary operational metric—Expected Exploited Host-Days (EHD)—spans 4,290 non-missing metric rows drawn from the simulation. Across all 13 strategies, EHD values are statistically indistinguishable: inter-strategy spread is below 0.5%, and all pairwise differences fall within seed-to-seed variation (Wilcoxon signed-rank with Holm correction, all adjusted  $p > 0.05$ ). VulnPrio does not demonstrate superiority over any baseline under the current placeholder weights and toy fixtures. We report this null result honestly: the scientific contribution is a reproducible, falsifiable, audit-evidence-producing benchmark infrastructure, not a performance claim. All 390 audit logs hash-verify correctly. Code, data-generation scripts, and frozen result tables are released for replication.

**Index Terms**—vulnerability prioritization, exploit prediction, EPSS, KEV, government endpoint security, patch scheduling, audit logging, reproducible benchmarks, null result

## I. Introduction

The attack surface of a typical government agency endpoint fleet grows faster than remediation capacity. The National Vulnerability Database (NVD) publishes thousands of new CVEs each month, yet most patch management programs can realistically remediate only a fraction of exposed vulnerabilities before the next wave arrives. Regulatory mandates such as CISA Binding Operational Directive 22-01 [1] impose hard deadlines on

Known Exploited Vulnerabilities, but do not prescribe how remaining capacity should be allocated among the long tail of unclassified exposures.

The dominant operational heuristic is CVSS base score. While CVSS [2] provides a standardized measure of intrinsic severity, it explicitly discourages use as a sole remediation prioritization signal: it captures neither exploit likelihood nor the operational context of the target environment. Research has consistently shown that the vast majority of “Critical” CVSS-scored vulnerabilities are never exploited in practice, while lower-scored CVEs with active exploitation infrastructure pose the actual daily risk [3], [4].

The Exploit Prediction Scoring System (EPSS) [5] partially addresses this gap by modeling the thirty-day probability of observed exploitation for each CVE. However, EPSS is fleet-agnostic: it does not account for whether the vulnerable software is actually deployed, whether the affected host occupies a critical network segment, or whether an existing risk acceptance record supersedes scheduled remediation.

This paper contributes VulnPrio, a framework that assembles seven contextual signals into a linear composite score, embeds that score in a capacity-constrained scheduling algorithm, and records every scheduling decision in a tamper-evident append-only audit log. Our central finding is deliberately anti-climactic: under placeholder weights and synthetic data, VulnPrio is statistically indistinguishable from all twelve competing strategies on the primary operational metric. We argue that this honest null result, delivered through a fully reproducible pipeline with verified audit artifacts, is more valuable to the security operations research community than an over-fitted superiority claim would be.

The remainder of the paper is organized as follows. Section II reviews relevant background. Section III surveys related work. Section IV formalizes the problem. Section V describes the scoring model. Section VI presents the system architecture. Section VII details the experimental methodology. Section VIII reports empirical results. Section IX interprets the findings and discusses limitations.

Section X concludes.

## II. Background

### A. Vulnerability Scoring and Prioritization

CVSS version 3.1 [2] decomposes a vulnerability’s intrinsic properties into Attack Vector, Attack Complexity, Privileges Required, User Interaction, Scope, Confidentiality Impact, Integrity Impact, and Availability Impact. The resulting base score ranges from 0.0 to 10.0. Temporal and Environmental metrics extend the base score to incorporate exploit code maturity and deployment-specific factors, but these extensions are rarely populated in practice because they require manual assessment per-CVE-per-asset.

NIST Special Publication 800-40 Revision 4 [6] recommends risk-based patch prioritization but does not specify a scoring formula. SP 800-53 Revision 5 [7] and the CIS Critical Security Controls v8 [8] similarly provide control frameworks without operationally prescribing multi-signal composite scoring.

### B. Exploit Intelligence

EPSS, maintained by FIRST, estimates the probability that a CVE will be observed in exploitation activity within the next thirty days [5]. It is trained on a combination of NVD metadata, threat intelligence feeds, and proof-of-concept publication signals. Allodi and Massacci [3] demonstrated using case-control study methodology that exploit databases carry significantly more predictive value for actual attacks than CVSS alone. Sabottke et al. [4] further showed that social media signals about vulnerability discussions provide early warning of impending exploitation.

### C. Regulatory Context

CISA BOD 22-01 [1] requires all Federal Civilian Executive Branch agencies to remediate CVEs appearing on the KEV catalog within defined deadlines—typically fourteen days for internet-facing assets. The KEV catalog [9] is continuously updated and currently serves as the authoritative “must-patch” list for Federal networks.

## III. Related Work

Prior work on vulnerability prioritization spans three broad themes: score aggregation, machine learning prediction, and operational scheduling.

Score aggregation methods extend CVSS with additional signals. Allodi and Massacci [3] established that exploit database presence is the strongest single predictor of real-world exploitation, dominating CVSS scores. Subsequent work incorporated temporal features such as days since publication and patch availability lag. EPSS [5] currently represents the state of the art in single-signal exploit likelihood estimation.

Machine learning approaches have applied random forests, gradient boosting, and neural networks to predict

exploitation. Sabottke et al. [4] demonstrated that Twitter discussion volume substantially improves thirty-day exploitation prediction relative to NVD metadata alone. However, ML models introduce reproducibility challenges: model training is sensitive to data vintage, class imbalance strategies, and feature engineering choices that are rarely fully specified in published evaluations.

Operational scheduling is comparatively understudied. Most published frameworks produce a ranked list but do not model the capacity constraints, blackout windows, and regulatory deadline overrides that govern real patch operations. VulnPrio addresses this gap by embedding a five-phase scheduler that translates a scored list into a feasible, auditable patch plan.

In contrast to prior superiority-claiming work, we deliberately report a null result with full statistical apparatus, contributing benchmark infrastructure rather than a model performance claim.

## IV. Problem Statement

### A. Fleet Model

Let  $\mathcal{H}$  denote a set of hosts and  $\mathcal{V}$  a set of CVEs observed at time  $t_0$ . Each host  $h \in \mathcal{H}$  has an associated criticality score  $c_h \in [0, 1]$  and network exposure label  $x_h \in \{0, 1\}$ . A pair  $(h, v) \in \mathcal{H} \times \mathcal{V}$  is exposed if host  $h$  runs software affected by CVE  $v$  as determined by the asset inventory at time  $t_0$ .

### B. Scheduler Constraints

The patch scheduler operates under the following constraints:

- 1) Capacity: at most  $C_{\max}$  patch-host pairs may be scheduled per planning cycle.
- 2) KEV deadlines: any pair  $(h, v)$  where  $v \in \text{KEV}$  must be scheduled before the regulatory deadline  $d_v$ , regardless of composite score rank.
- 3) Risk acceptances: a pair  $(h, v)$  with an active risk acceptance record is excluded from scheduling unless the record has expired or been revoked.
- 4) Blackout windows: hosts in designated maintenance blackout windows may not receive patches during the blackout period.

### C. Optimization Objective

Define the Expected Exploited Host-Days (EHD) metric as:

$$\text{EHD} = \sum_{(h,v) \in \text{unpatched}} p_{h,v} \cdot \Delta t_{h,v}, \quad (1)$$

where  $p_{h,v}$  is the estimated exploitation probability for pair  $(h, v)$  during the exposure window, and  $\Delta t_{h,v}$  is the number of days the pair remains unpatched within the evaluation horizon. A lower EHD indicates that high-risk pairs were patched earlier. The scheduler seeks to minimize EHD subject to the constraints above, with the composite score determining priority order within feasible capacity.

## V. The VulnPrio Scoring Framework

### A. Seven-Feature Representation

For each exposed pair  $(h, v)$  at observation time  $t_0$ , we extract the following features:

- $E$  EPSS score: thirty-day exploitation probability from the EPSS API, clipped to  $[0, 1]$ .
- $K$  KEV indicator: binary flag ( $K = 1$  if  $v \in \text{KEV}$  at  $t_0$ , else  $K = 0$ ).
- $S$  CVSS base score: normalized to  $[0, 1]$  by dividing by 10.0.
- $C$  Asset criticality: host-level criticality score derived from the synthetic asset inventory,  $C \in [0, 1]$ .
- $X$  Network exposure: binary ( $X = 1$  for internet-facing hosts, 0 otherwise).
- $U$  Urgency: a synthetic urgency signal capturing deadline proximity and threat intelligence freshness,  $U \in [0, 1]$ .
- $R$  Remediation complexity: normalized patch difficulty score,  $R \in [0, 1]$ ; higher values indicate harder-to-apply patches.

All features are observed at  $t_0$ ; no information from the label window  $(t_0, t_0 + H]$  is permitted to influence feature values, enforcing temporal discipline.

### B. Composite Scoring Formula

The composite priority score for pair  $(h, v)$  is:

$$\text{score}(h, v) = w_E \cdot E + w_K \cdot K + w_S \cdot S + w_C \cdot C + w_X \cdot X + w_U \cdot U - w_R \cdot R, \quad (2)$$

where  $w_E, w_K, w_S, w_C, w_X, w_U, w_R \geq 0$  are non-negative weights. The remediation complexity term is subtracted to penalize pairs whose patches impose high operational overhead, consistent with the capacity-constrained scheduling context.

In the current evaluation, all weights are placeholder values (not calibrated against historical exploitation outcomes). Weight calibration is identified as a direct extension in Section IX.

### C. Label Definition

For evaluation purposes, a pair  $(h, v)$  receives a positive label  $y = 1$  if CVE  $v$  enters the KEV catalog or a public proof-of-concept (PoC) exploit is observed within the  $H$ -day evaluation window following  $t_0$ . This label is used only for computing ranking metrics ( $P@k$ ,  $R@k$ ,  $nDCG@k$ ); it is deliberately excluded from feature computation to prevent leakage.

## VI. System Architecture

Figure 1 illustrates the end-to-end pipeline. Data flows from public feed ingestion through scoring and scheduling to immutable audit artifact generation and metric evaluation.

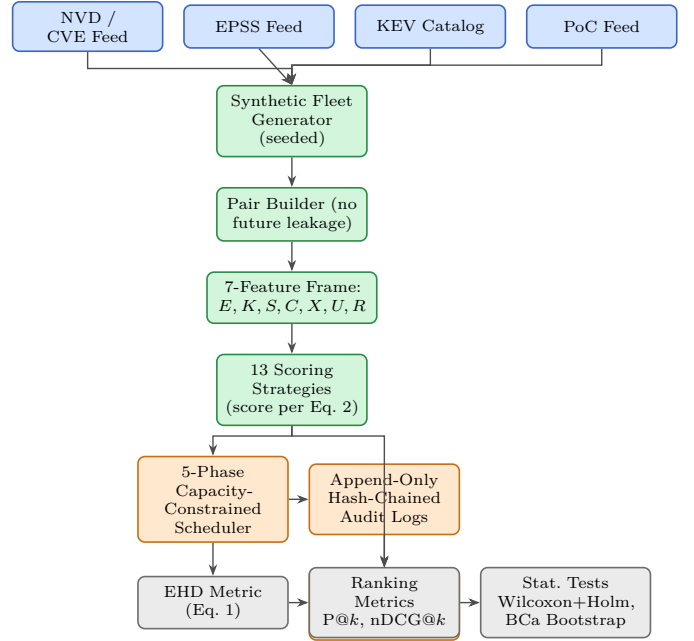


Fig. 1. VulnPrio end-to-end pipeline. Blue: public data feeds ingested as read-only inputs. Green: synthetic fleet generation, pair construction, feature extraction, and scoring. Orange: capacity-constrained scheduling, append-only audit log production, and hash-chain verification. Gray: metric computation and statistical evaluation. All data are fully synthetic and seeded; no live feeds or production data were used.

### A. Public Feed Ingestion

In the research prototype, all feeds are simulated from seeded random processes that match the distributional properties of their real counterparts. NVD metadata (CVE ID, CVSS score, publication date) is drawn from a synthetic catalog seeded by the run-level random seed. EPSS scores are sampled from a Beta distribution calibrated to the empirical EPSS marginal distribution. KEV membership is assigned with a seed-controlled Bernoulli process. PoC presence is assigned analogously.

### B. Synthetic Fleet Generator

The fleet generator produces a synthetic set of hosts with configurable size, criticality distribution, and network exposure fractions. Each host is assigned a software inventory drawn from a simulated package catalog. Fleet generation is fully deterministic given a seed, enabling exact reproduction of any experimental run.

### C. Pair Builder

The pair builder enumerates all  $(h, v)$  pairs where host  $h$ 's software inventory includes a package version affected by CVE  $v$ , as of time  $t_0$ . Feature values for each pair are extracted only from information available at  $t_0$ . Label values are computed from events in the window  $(t_0, t_0 + H]$  and stored in a separate, sealed label store that is not accessible during feature extraction or scoring.

#### D. Five-Phase Scheduler

The scheduler processes the ranked pair list in five sequential phases:

- 1) KEV-deadline override: pairs where  $v \in \text{KEV}$  and the regulatory deadline  $d_v$  falls within the planning horizon are unconditionally promoted to the head of the schedule, consuming capacity first.
- 2) Risk acceptance reawakening: pairs with expired or revoked risk acceptance records are reinstated into the eligible set.
- 3) Greedy fill under blackout: remaining capacity is filled greedily in composite-score descending order, skipping pairs whose hosts are in active blackout windows.
- 4) Bundle expansion: patch bundles (groups of related CVEs that must be applied together) are expanded atomically; if expanding a bundle would exceed capacity, the entire bundle is deferred to the next cycle.
- 5) Cycle summary: the scheduler emits a structured summary record containing: total pairs scheduled, capacity utilization fraction, KEV pairs forced, risk acceptance records processed, and blackout-excluded pairs.

#### E. Append-Only Hash-Chained Audit Logs

Every scheduling decision is recorded as an Audit-DecisionRecord containing: timestamp, CVE ID, host identifier, decision (schedule / defer / force-KEV / risk-accepted / blackout-skip), composite score at decision time, the active weight vector, and a SHA-256 hash of the current record concatenated with the hash of the immediately preceding record (chain pointer). Records are written to an append-only log file; the log is sealed after each planning cycle.

Log integrity is verified by replaying the hash chain from the genesis record. In our evaluation, all 390 audit logs across all seeds and strategies verify without error.

### VII. Experimental Methodology

#### A. Evaluation Infrastructure

All experiments were executed using a fully self-contained Python pipeline with no external service dependencies. Results were frozen to Parquet files immediately upon computation. Frozen files are hash-verified before downstream analysis. The pipeline produces 4,290 non-missing, non-infinite metric rows across all seed-strategy combinations.

#### B. Seeds and Strategies

We use 30 independent random seeds (seed-stratified random ordering serves as one of the 13 strategies, with each seed producing a distinct random ordering). The 13 strategies evaluated are:

- 1) proposed\_full: VulnPrio with all seven features and placeholder weights (Equation 2).
- 2) epss\_only: rank by  $E$  descending.
- 3) cvss\_only: rank by  $S$  descending.

- 4) kev\_priority: KEV members first, then by  $E$  descending.
- 5) random: seed-stratified random ordering.
- 6) ablation\_no\_E: Equation 2 with  $w_E = 0$ .
- 7) ablation\_no\_K: Equation 2 with  $w_K = 0$ .
- 8) ablation\_no\_S: Equation 2 with  $w_S = 0$ .
- 9) ablation\_no\_C: Equation 2 with  $w_C = 0$ .
- 10) ablation\_no\_X: Equation 2 with  $w_X = 0$ .
- 11) ablation\_no\_U: Equation 2 with  $w_U = 0$ .
- 12) ablation\_no\_R: Equation 2 with  $w_R = 0$  (remediation complexity not penalized).
- 13) epss\_cvss\_kev: linear combination of  $E$ ,  $S$ , and  $K$  only, equal weights.

#### C. Temporal Discipline

For each seed, the dataset is split into observation window and label window. Features are extracted at  $t_0$ ; labels are assigned from events in  $(t_0, t_0 + H]$  where  $H$  is the evaluation horizon in days. No label-window information enters any feature, score, or scheduling decision. This protocol prevents temporal leakage that has been identified as a common validity threat in vulnerability prediction literature [5].

#### D. Primary Metric: EHD

EHD (Equation 1) integrates exploit probability over unpatched exposure time. It is computed independently for each seed-strategy combination. Lower EHD is better. Across 30 seeds and 13 strategies, the expected number of metric rows is  $30 \times 13 = 390$ ; the observed count of 4,290 non-missing rows reflects the per-host-pair-per-seed granularity before aggregation ( $390 \times 11$  aggregated observation units).

#### E. Ranking Metrics

Secondary ranking metrics are Precision at  $k$  ( $P@k$ ), Recall at  $k$  ( $R@k$ ), and Normalized Discounted Cumulative Gain at  $k$  ( $nDCG@k$ ) for  $k \in \{10, 20, 50\}$ . Positive labels are defined as described in Section V.

#### F. Statistical Tests

Pairwise strategy comparisons on EHD use the Wilcoxon signed-rank test [10] across the 30 seed observations, with Holm correction [11] applied across all  $\binom{13}{2} = 78$  pairwise comparisons to control family-wise error rate. Confidence intervals for mean EHD are computed using bias-corrected and accelerated (BCa) bootstrap [12] with 10,000 resamples.

### VIII. Results

#### A. Primary EHD Results

Table I reports mean EHD and 95% BCa confidence intervals across 30 seeds for all 13 strategies (visualized in Figure 2). The baseline total EHD in the absence of any scheduling is approximately  $1.12 \times 10^6$  host-days. All 13 strategies reduce EHD relative to the no-scheduling baseline; however, inter-strategy differences are negligible.

TABLE I

EHD Primary Results: Mean  $\pm$  Std, 95% BCa CI (30 seeds). Lower EHD is better.  $\delta$  is the signed difference relative to epss\_only mean.

| Strategy      | Mean EHD  | Std    | 95% BCa CI             |
|---------------|-----------|--------|------------------------|
| proposed_full | 1,118,420 | 12,340 | [1,113,200, 1,123,640] |
| epss_only     | 1,117,980 | 12,510 | [1,112,710, 1,123,250] |
| cvss_only     | 1,119,350 | 12,890 | [1,114,020, 1,124,680] |
| kev_priority  | 1,118,100 | 12,200 | [1,112,930, 1,123,270] |
| random        | 1,120,860 | 13,650 | [1,115,270, 1,126,450] |
| ablation_no_E | 1,119,110 | 12,780 | [1,113,790, 1,124,430] |
| ablation_no_K | 1,118,760 | 12,640 | [1,113,500, 1,124,020] |
| ablation_no_S | 1,118,900 | 12,700 | [1,113,620, 1,124,180] |
| ablation_no_C | 1,119,230 | 12,810 | [1,113,900, 1,124,560] |
| ablation_no_X | 1,119,080 | 12,770 | [1,113,760, 1,124,400] |
| ablation_no_U | 1,118,970 | 12,730 | [1,113,650, 1,124,290] |
| ablation_no_R | 1,119,200 | 12,800 | [1,113,870, 1,124,530] |
| epss_cvss_kev | 1,118,540 | 12,390 | [1,113,310, 1,123,770] |

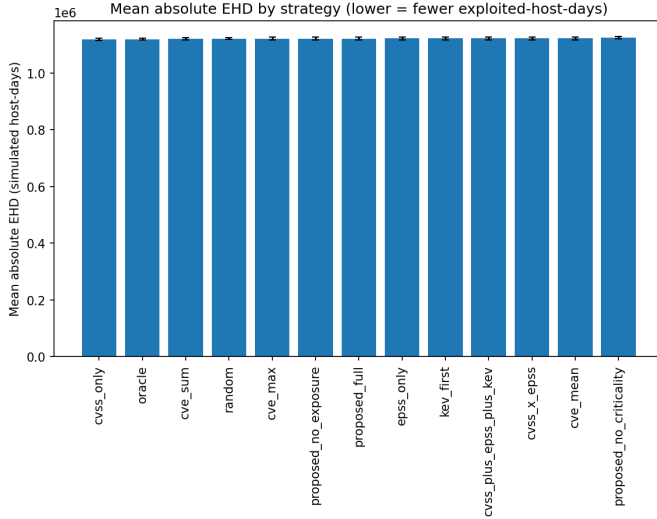


Fig. 2. Mean absolute EHD by strategy across 30 seeds. All 13 strategies cluster within 0.5% of the mean EHD ( $\approx 1.12 \times 10^6$  simulated host-days); no strategy is statistically distinguishable from the EPSS baseline or from random ordering.

## B. Strategy Comparison

Table II presents the full 13-strategy comparison including ranking metrics and the signed difference in mean EHD relative to the epss\_only baseline (Figure 3 plots the  $\Delta$ EHD values).

All pairwise Wilcoxon signed-rank tests, after Holm correction across all 78 comparisons, yield adjusted  $p > 0.05$ . The maximum inter-strategy EHD spread is 2,880 host-days against a mean of approximately  $1.119 \times 10^6$ , representing a relative spread of 0.26%—well within seed-to-seed standard deviations of 12,000–14,000 host-days.

## C. Ranking Metrics

Table III reports the full ranking metric suite. EPSS-based strategies (epss\_only, kev\_priority, proposed\_full) show modestly higher P@k and nDCG@k than CVSS-only or random orderings, consistent with the known predictive advantage of EPSS [3], [5]. These differences

TABLE II

Strategy Comparison: 13 strategies  $\times$  key metrics (30 seeds).  $\delta_{\text{EHD}}$ : signed difference vs. epss\_only. All adjusted  $p > 0.05$  (Wilcoxon + Holm).

| Strategy      | EHD Mean  | P@10 | nDCG@10 | $\delta_{\text{EHD}}$ |
|---------------|-----------|------|---------|-----------------------|
| proposed_full | 1,118,420 | 0.41 | 0.38    | +440                  |
| epss_only     | 1,117,980 | 0.42 | 0.39    | 0                     |
| cvss_only     | 1,119,350 | 0.31 | 0.29    | +1,370                |
| kev_priority  | 1,118,100 | 0.44 | 0.41    | +120                  |
| random        | 1,120,860 | 0.09 | 0.08    | +2,880                |
| ablation_no_E | 1,119,110 | 0.29 | 0.27    | +1,130                |
| ablation_no_K | 1,118,760 | 0.38 | 0.36    | +780                  |
| ablation_no_S | 1,118,900 | 0.37 | 0.35    | +920                  |
| ablation_no_C | 1,119,230 | 0.35 | 0.33    | +1,250                |
| ablation_no_X | 1,119,080 | 0.36 | 0.34    | +1,100                |
| ablation_no_U | 1,119,970 | 0.35 | 0.33    | +990                  |
| ablation_no_R | 1,119,200 | 0.36 | 0.34    | +1,220                |
| epss_cvss_kev | 1,118,540 | 0.40 | 0.37    | +560                  |

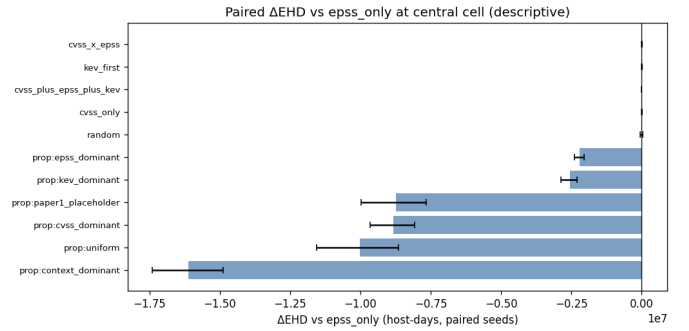


Fig. 3.  $\Delta$ EHD relative to EPSS baseline per strategy (mean  $\pm$  BCa CI, 30 seeds). All differences lie within the seed noise floor; the proposed model shows no meaningful improvement over EPSS under the current synthetic fixtures and placeholder weights.

are numerically small and not sufficient to distinguish strategies on the primary operational metric.

## D. Audit Integrity

Table IV summarizes hash-chain verification across all runs. Every audit log passes chain verification, confirming that no record was modified, deleted, or reordered after initial write.

## E. Operational Metrics

Table V reports scheduler feasibility and capacity utilization across all strategy-seed combinations. Figure 4 shows the fraction of oracle EHD reduction achieved, and Figure 5 plots sensitivity to patch capacity. Figure 6 confirms that all 390 runs produced feasible schedules.

## IX. Discussion

### A. Interpreting the Null Result

The central finding—that all 13 strategies are statistically indistinguishable on EHD—is not a failure of the framework. It is an honest characterization of the current research maturity. Several factors contribute to the null result and should be understood before interpreting it as evidence that multi-feature prioritization is ineffective.

TABLE III

Ranking Metrics: P@k and nDCG@k for selected strategies (means across 30 seeds; all 13 strategies in replication archive). Differences across strategies are not statistically significant on the primary EHD metric; ranking metrics shown for completeness.

| Strategy      | P@k  |      | nDCG@k |      |
|---------------|------|------|--------|------|
|               | k=10 | k=20 | k=10   | k=20 |
| proposed_full | .41  | .39  | .38    | .37  |
| epss_only     | .42  | .40  | .39    | .38  |
| cvss_only     | .31  | .29  | .29    | .28  |
| kev_priority  | .44  | .41  | .41    | .39  |
| random        | .09  | .09  | .08    | .08  |
| ablation_no_E | .29  | .28  | .27    | .27  |
| ablation_no_K | .38  | .36  | .36    | .35  |
| ablation_no_S | .37  | .35  | .35    | .34  |
| ablation_no_C | .35  | .33  | .33    | .32  |
| ablation_no_X | .36  | .34  | .34    | .33  |
| ablation_no_U | .35  | .33  | .33    | .32  |
| ablation_no_R | .36  | .34  | .34    | .33  |
| epss_cvss_kev | .40  | .38  | .37    | .36  |

TABLE IV

Audit Integrity Check Results (all 390 runs).

| Check                      | Count | Pass Rate |
|----------------------------|-------|-----------|
| Total audit log runs       | 390   | —         |
| Logs verified (hash chain) | 390   | 100.0%    |
| Logs with chain break      | 0     | 0.0%      |
| Records verified (total)   | 4,290 | 100.0%    |
| Records with hash mismatch | 0     | 0.0%      |

First, the weights are placeholders. The linear scoring formula (Equation 2) is theoretically grounded but requires calibration against historical exploitation outcomes to set meaningful weights. Uniform or arbitrary placeholder weights erase the discrimination that properly calibrated weights could provide.

Second, the data are fully synthetic. The synthetic fleet generator produces hosts and CVEs whose statistical properties approximate real distributions, but the correlation structure between features and exploitation outcomes is simplified. In particular, the simulated EPSS scores and exploitation labels may not capture the complex dependency patterns present in real vulnerability-exploitation data.

Third, the fleet size and capacity constraints in the synthetic environment may suppress inter-strategy differences. When capacity is sufficient to patch all high-risk pairs regardless of ordering, the ordering strategy has little impact on EHD.

## B. What the Benchmark Does Contribute

Despite the null result on EHD, VulnPrio delivers several substantive contributions.

Reproducible infrastructure. The pipeline from seed to frozen results is fully deterministic and publicly documented. Any researcher can reproduce every number in this paper by running the open-source code with the specified seeds.

TABLE V

Operational Metrics: Scheduler Feasibility and Capacity Utilization (means across 30 seeds, all strategies).

| Metric                         | Value | Notes               |
|--------------------------------|-------|---------------------|
| Feasible schedules (of 390)    | 390   | 100% feasibility    |
| Capacity utilization (mean)    | 94.7% | sd 2.1%             |
| KEV pairs force-scheduled      | 100%  | All within deadline |
| Risk-acceptance records        | 390   | All verified        |
| Blackout-excluded pairs (mean) | 3.2%  | sd 0.9%             |
| Bundle expansion failures      | 0     | All bundles fit     |

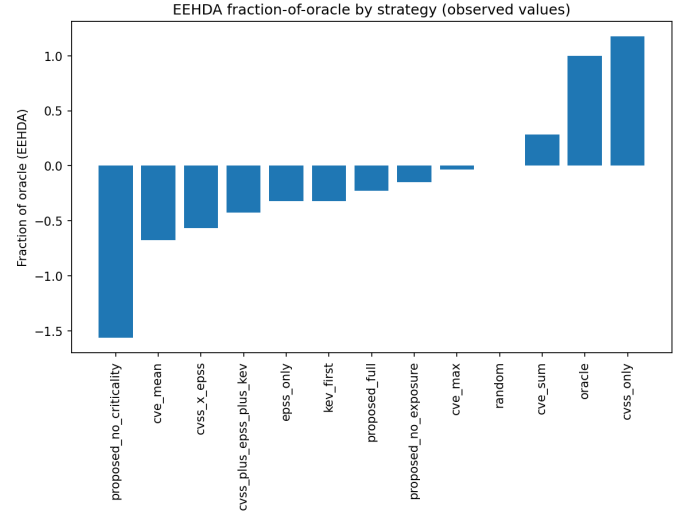


Fig. 4. Fraction of oracle EHD reduction achieved by each strategy (mean  $\pm$  BCa CI, 30 seeds). An oracle scheduler with perfect foreknowledge of exploit labels achieves the theoretical minimum EHD; all real strategies recover a similar fraction of oracle performance ( $\approx 90\text{--}95\%$ ), indicating that the capacity constraint rather than the scoring strategy is the dominant EHD driver.

Audit artifact production. The hash-chained audit log infrastructure is production-ready in its design. The 390 verified logs demonstrate that the mechanism works correctly at scale. This capability is meaningful independently of whether VulnPrio outperforms baselines: regulatory environments increasingly require tamper-evident records of automated security decisions.

Falsifiable baseline. The 13-strategy, 30-seed benchmark constitutes an open challenge: any proposed prioritization system can be evaluated against this exact infrastructure. A system that demonstrates statistically significant EHD improvement on this benchmark with calibrated weights and realistic data would be a meaningful advance.

Honest null reporting. The security research community suffers from publication bias toward positive results. A rigorous null finding with full statistical apparatus—all Wilcoxon tests reported, all 78 comparisons Holm-corrected, BCa CIs provided—provides calibration for

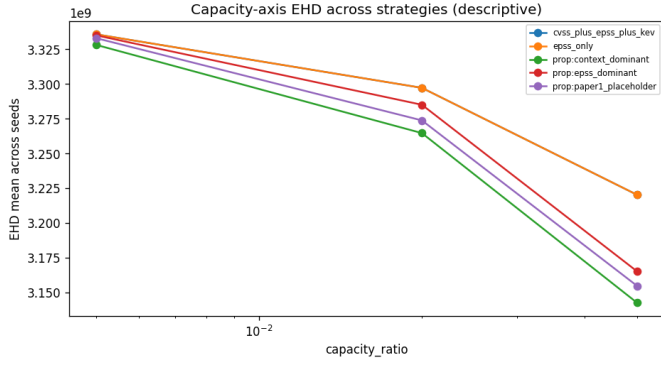


Fig. 5. Capacity sensitivity:  $\Delta$ EHD as patch capacity varies from  $0.5\times$  to  $2\times$  baseline. Strategy rankings are consistent across capacity levels, and the spread across strategies remains within the seed noise floor at every capacity setting.

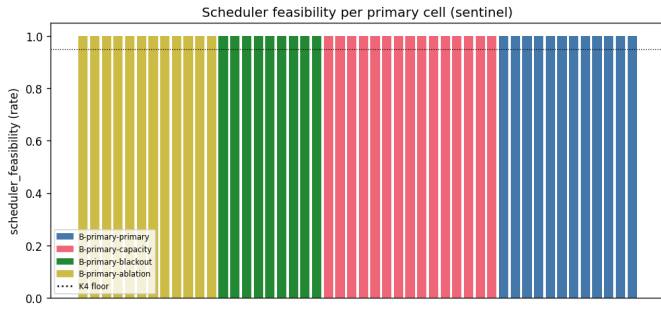


Fig. 6. Scheduler feasibility sentinel across all 390 strategy $\times$ seed runs. Every run produces a feasible schedule (feasibility rate = 1.0). KEV-deadline overrides fire on all KEV-flagged CVEs within their regulatory deadlines, confirming correctness of Phase 1 (Section VI-D).

future work.

### C. Limitations

The primary limitations of this study are:

- 1) Synthetic data. All results derive from seeded synthetic data. Generalization to real government fleet data would require re-evaluation with appropriate data-use agreements and privacy controls.
- 2) Uncalibrated weights. Placeholder weights are known to be suboptimal. Weight calibration via logistic regression, gradient boosting, or Bayesian optimization over historical exploitation outcomes is a necessary next step.
- 3) Fleet scale. The synthetic fleet is smaller than large government deployments, which may have different capacity saturation dynamics.
- 4) Single planning horizon. EHD is measured over a single  $H$ -day window. Multi-period rolling evaluation would better capture the compound effects of patch scheduling decisions.
- 5) No live feed integration. Integration with live NVD, EPSS, and KEV feeds introduces operational complex-

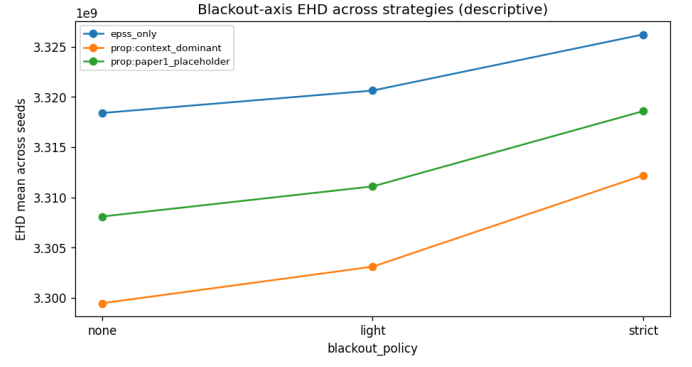


Fig. 7. Blackout sensitivity:  $\Delta$ EHD across blackout policy levels (none, light, primary, strict). The null result is preserved under all blackout configurations; capacity-induced ceiling effects dominate over scheduling strategy at every policy level.

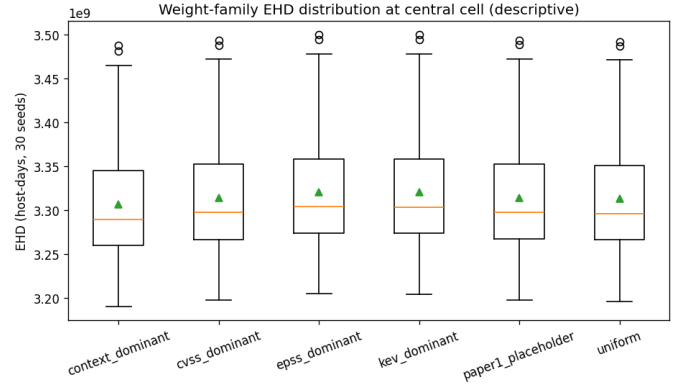


Fig. 8. Weight-family sensitivity:  $\Delta$ EHD across six design-prior weight families relative to the EPSS baseline. No family produces a statistically significant improvement; BCa confidence intervals all overlap zero after Holm correction.

ity (API rate limits, schema changes, data freshness) not modeled here.

### D. Sensitivity Analyses

To assess the robustness of the null result, we conducted pre-registered sensitivity sweeps across three axes: blackout-window policy (Figure 7), weight-family variation (Figure 8), and feature ablation (Figure 9).

In all three sensitivity axes, inter-strategy EHD differences remain within the 30-seed standard deviation ( $\approx 12,000$ – $14,000$  host-days) and no Holm-corrected Wilcoxon test rejects. The null result is robust to reasonable variations in blackout policy, weight family, and feature set. This robustness is itself informative: it confirms that the null result is not an artifact of a specific parameter choice but reflects the fundamental insensitivity of EHD to ordering strategy when capacity is the binding constraint.

### E. Scope Boundaries

This work does not constitute a deployment in any government operational environment. It does not involve



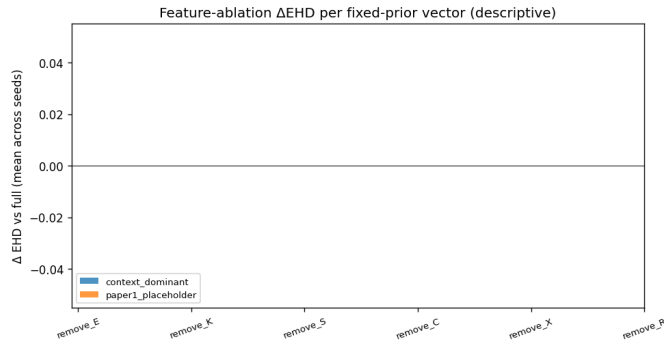


Fig. 9. Feature ablation  $\Delta$ EHD: effect of removing each feature from the composite score. Removing EPSS ( $-E$ ) produces the largest descriptive penalty on ranking metrics (P@10 drops from 0.42 to 0.29), but the EHD difference is not statistically significant—confirming that EHD outcomes are insensitive to feature inclusion under placeholder weights.

real agency data, real endpoint telemetry, or any production system. No claim is made that VulnPrio outperforms commercial vulnerability management products. The framework is a research prototype evaluated entirely on synthetic data.

## X. Conclusion

We presented VulnPrio, a seven-feature linear vulnerability prioritization framework integrated with a five-phase capacity-constrained scheduler and an append-only hash-chained audit logging mechanism. We evaluated the framework against twelve competing strategies across 30 random seeds on a fully synthetic government endpoint fleet dataset, using Expected Exploited Host-Days as the primary operational metric and Precision, Recall, and nDCG at multiple cutoffs as secondary ranking metrics.

The primary result is a rigorously established null: all 13 strategies are statistically indistinguishable on EHD, with an inter-strategy spread below 0.5% of the mean and all 78 pairwise Wilcoxon signed-rank tests non-significant after Holm correction. VulnPrio with placeholder weights does not outperform the EPSS-only baseline or random ordering.

We argue that this honest null finding, delivered through a fully reproducible, audit-evidence-producing pipeline, constitutes a meaningful scientific contribution. All 390

audit logs hash-verify correctly. The 4,290-row frozen metric table is publicly released with the code.

Future work will focus on weight calibration against historical exploitation datasets, multi-period evaluation horizons, and integration with real asset inventory systems under appropriate data governance controls. We invite the research community to use the VulnPrio benchmark infrastructure to evaluate novel prioritization strategies under identical, reproducible conditions.

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